Question:
There exist real numbers x and y, both greater than 1, such that $\log_x\left(y^x\right)=\log_y\left(x^{4y}\right)=10$. Find xy.

Reasoning + Answer:

Let's solve the problem step by step.

We are given that:

$$
\log_x(y^x) = 10 \quad \text{ and } \quad \log_y(x^{4y}) = 10
$$

Both $x > 1$ and $y > 1$, and we are to find xy.

Step 1: Simplify the first equation

$$
\log_x(y^x) = 10
$$

Use the logarithmic identity: $\log_b(a^c) = c \log_b a$

$$
\log_x(y^x) = x \log_x y = 10
$$

So:
$$
x \log_x y = 10 \quad \text{(Equation 1)}
$$

Step 2: Simplify the second equation

$$
\log_y(x^{4y}) = 10
$$

Similarly:

$$
\log_y(x^{4y}) = 4y \log_y x = 10
$$

So:
$$
4y \log_y x = 10 \quad \text{(Equation 2)}
$$

Step 3: Use change of base

We can express both logarithms in terms of natural logarithm, but more importantly, note that:

We know:
$$
\log_x y = \frac{1}{\log_y x}
$$

LetâĢĻs define:
$$
a = \log_x y
\quad \text{ and } \quad
b = \log_y x = \frac{1}{a}
$$

From Equation 1:
$$
x a = 10 \quad \Rightarrow \quad a = \frac{10}{x} \quad \text{(1)}
$$

From Equation 2:
$$
4y \cdot b = 10 \quad \text{ and since } b = \frac{1}{a}, \text{ then }
4y \cdot \frac{1}{a} = 10
$$
$$
\frac{4y}{a} = 10 \quad \text{(2)}
$$

Now substitute $a = \frac{10}{x}$ into Equation (2):

$$
\frac{4y}{10/x} = 10
$$
$$
4y \cdot \frac{x}{10} = 10
$$
$$
\frac{4xy}{10} = 10
$$
$$
\frac{2xy}{5} = 10
$$

Multiply both sides by 5:
$$
2xy = 50
$$

Divide by 2:
$$
xy = 25
$$

<answer>25</answer>